



Neutrinos via Charm Decays in Astrophysical Sources

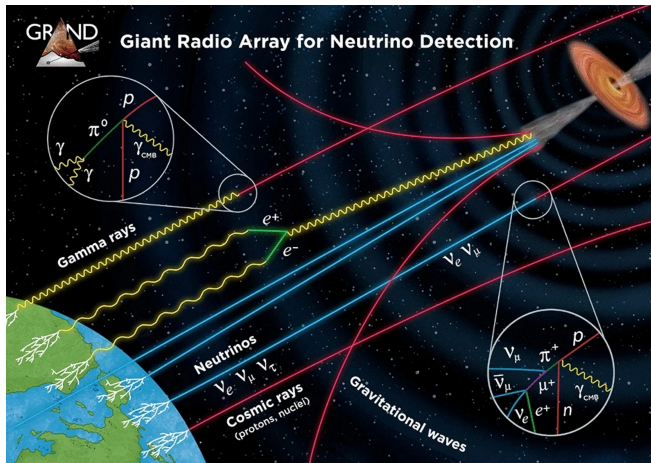
TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

Fritz Ali Agildere

September 3, 2024

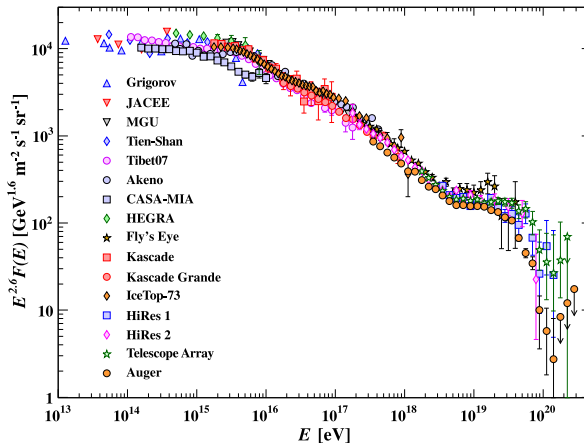
Technical University of Dortmund & Ruhr University Bochum

Multimessenger Astrophysics



X. L. Qian et al.
Nucl. Sci. Tech. (2021)

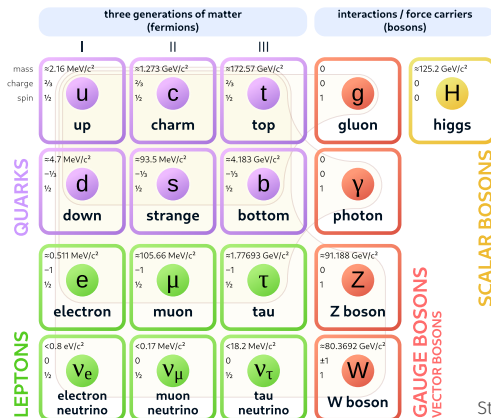
Cosmic Ray Spectrum & Cutoff



K. A. Olive et al.
Chin. Phys. C (2014)

Elementary Particles & Hadrons

Standard Model of Elementary Particles

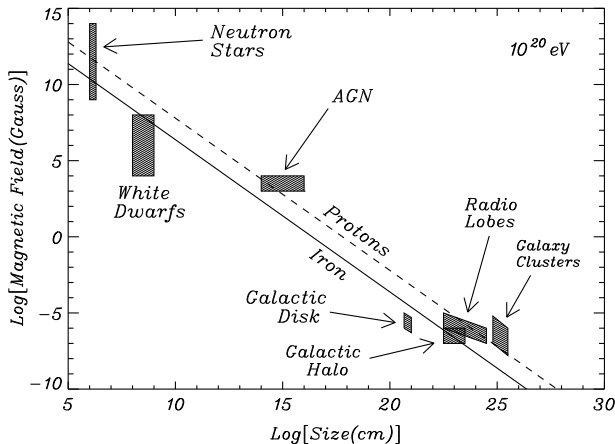


Standard Model
Wikimedia Commons (2019)

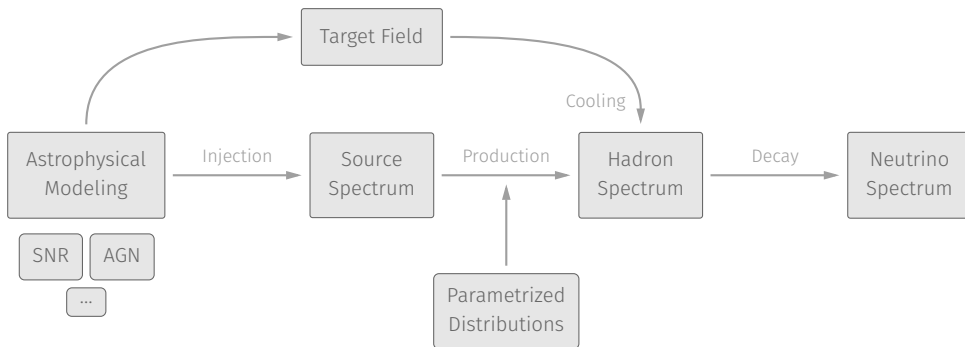
Relevant Hadron Properties

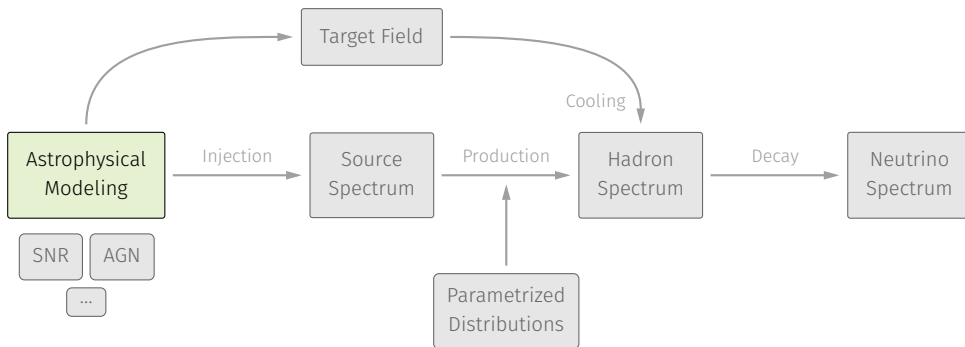
h	τ_h / s	m_h / GeV	s_h / GeV^2	$F_{h \rightarrow \nu}$
p	∞	0.94	15.98	0.000
π^+	26.0×10^{-9}	0.14	10.23	1.000
K^+	12.4×10^{-9}	0.49	12.62	0.636
D_0	0.4×10^{-12}	1.86	0.45	0.067
D^+	1.0×10^{-12}	1.87	0.40	0.176
D_s^+	0.5×10^{-12}	1.97	0.71	0.065
Λ_c^+	0.2×10^{-12}	2.29	1.61	0.045

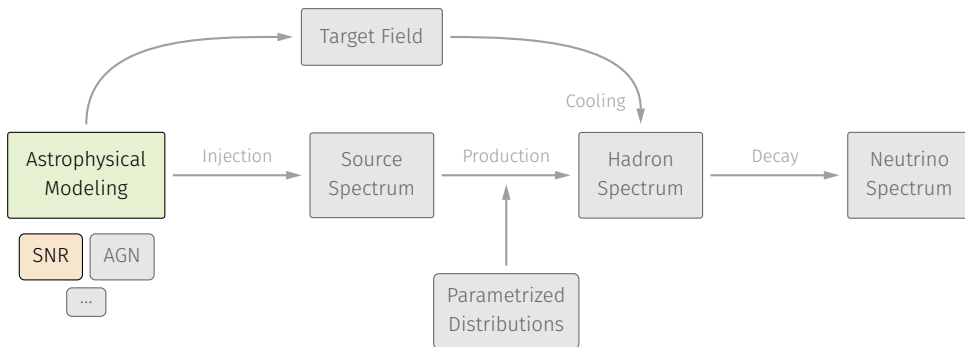
Plausible Cosmic Ray Origins

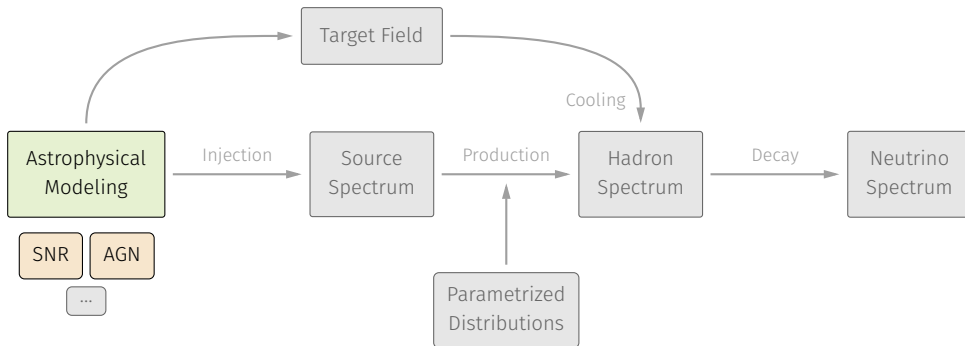


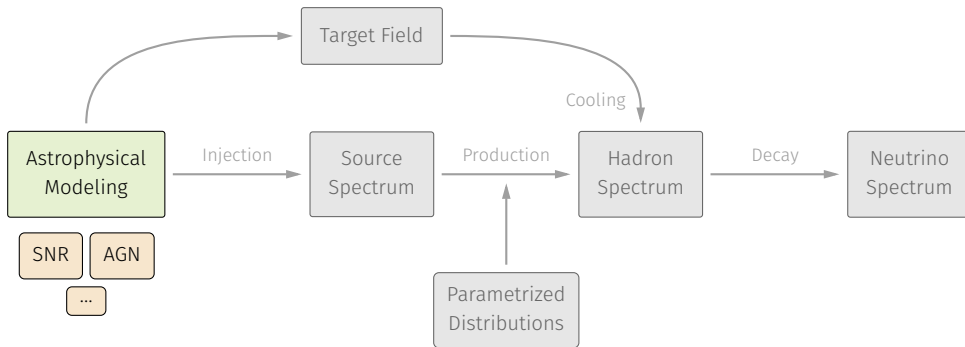
A. V. Olinto
AIP Conf. Proc. (2000)



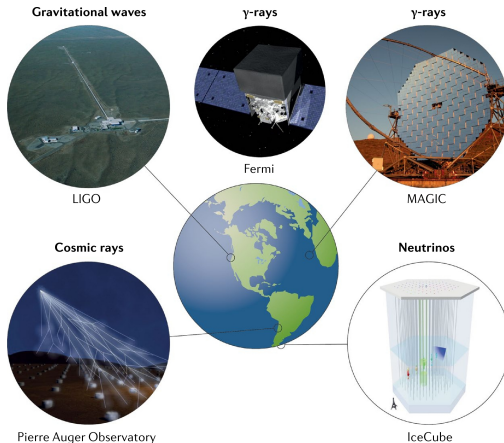






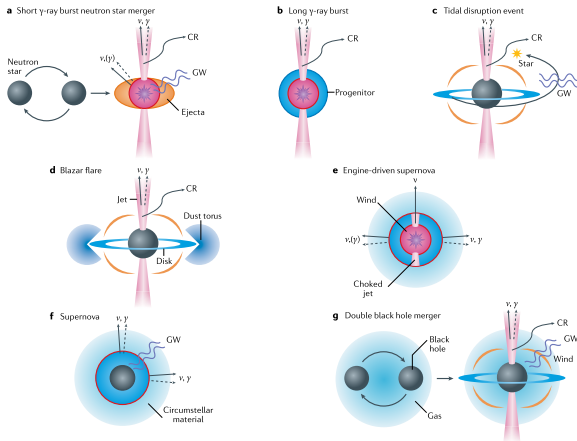


Different Messenger Types



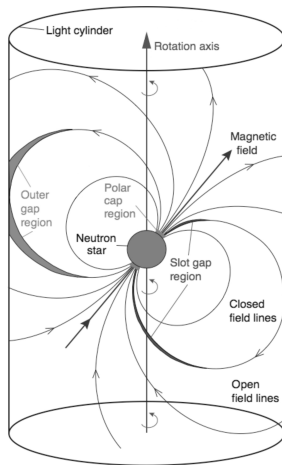
P. Mészáros et al.
Nat. Rev. Phys. (2019)

Source Regions & Events



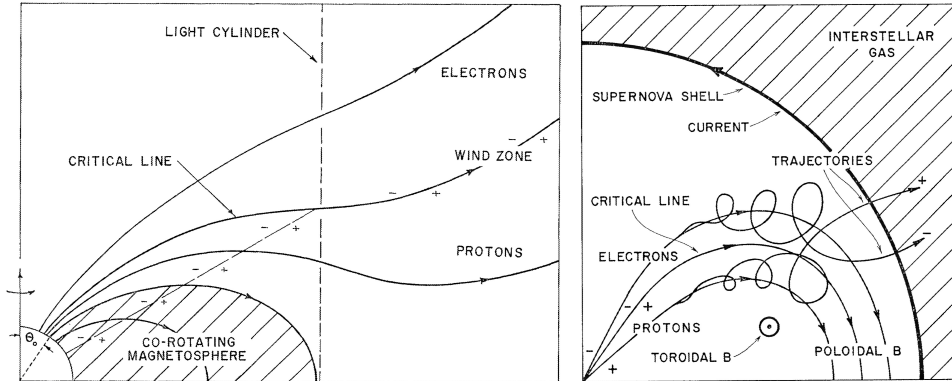
P. Mészáros et al.
Nat. Rev. Phys. (2019)

Pulsar



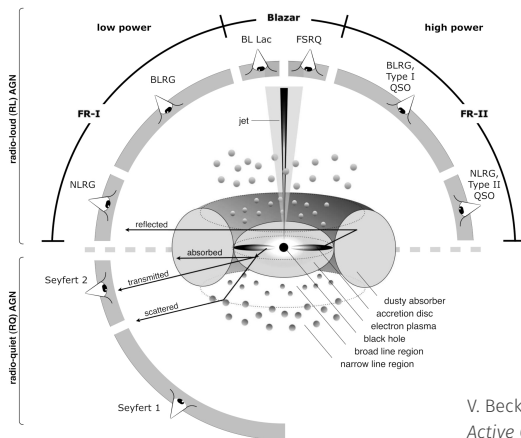
E. Aliu et al.
Science (2008)

Pulsar Particle Acceleration

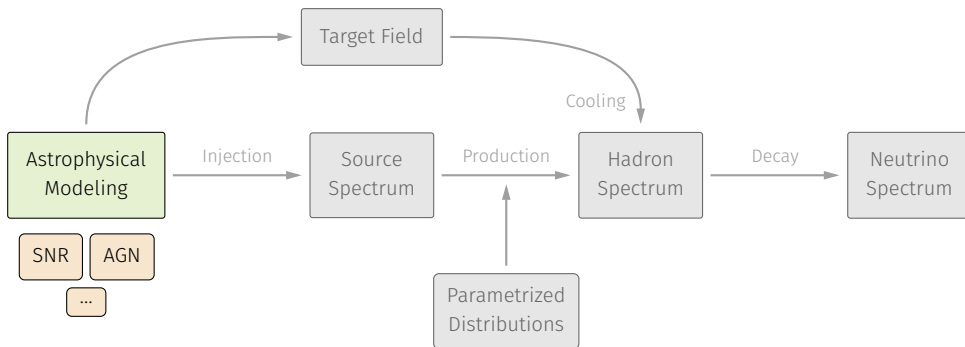


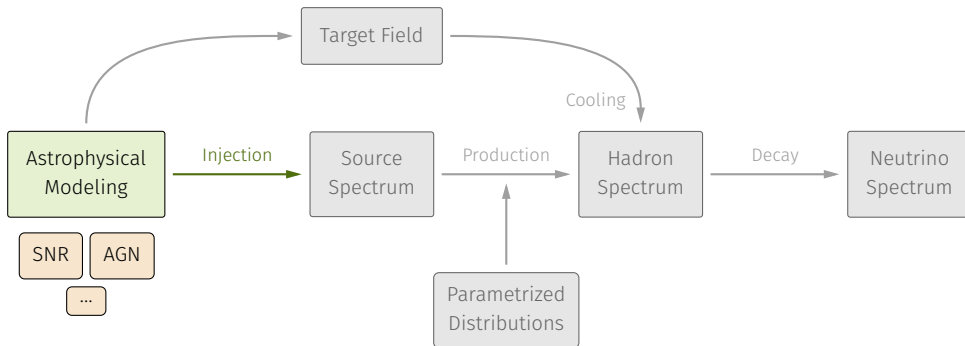
P. Goldreich, W. H. Julian *ApJ* (1969)

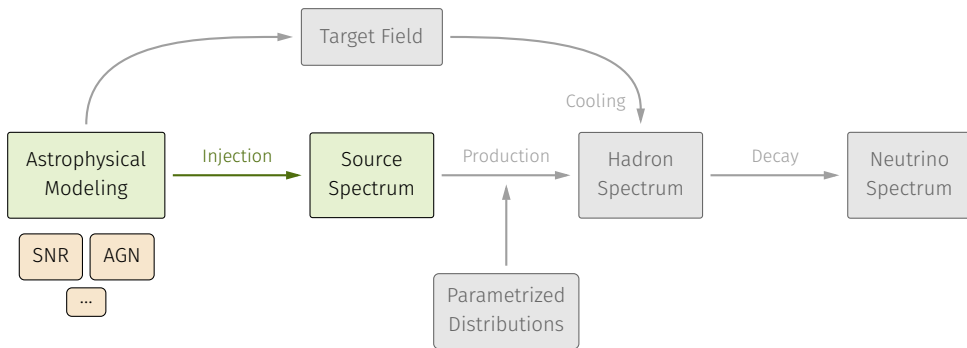
Unified AGN Classification Model



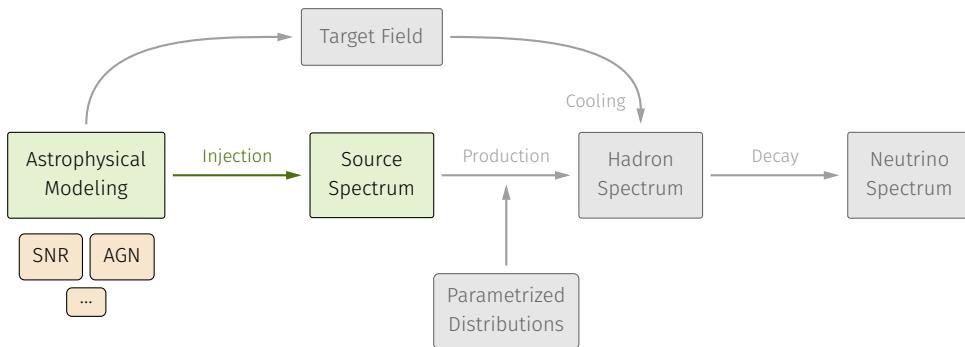
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

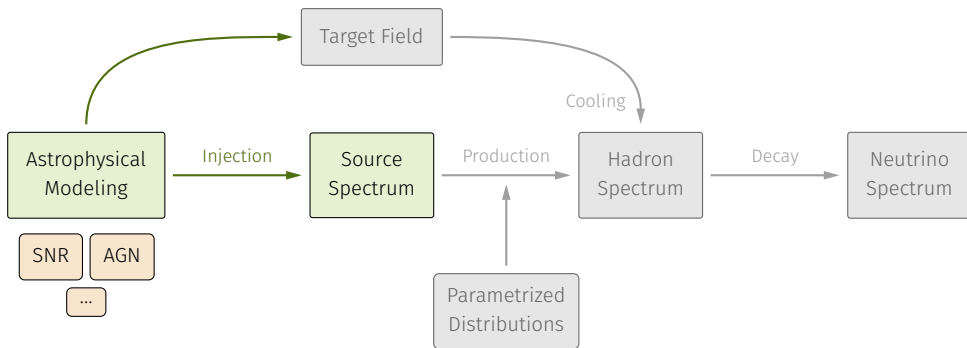


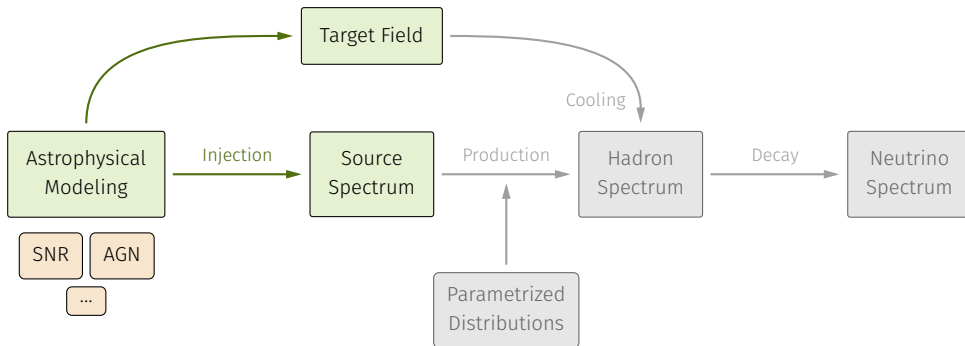




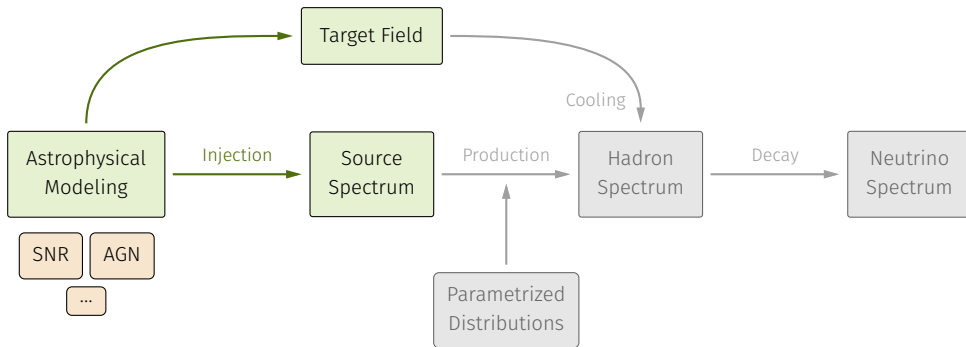


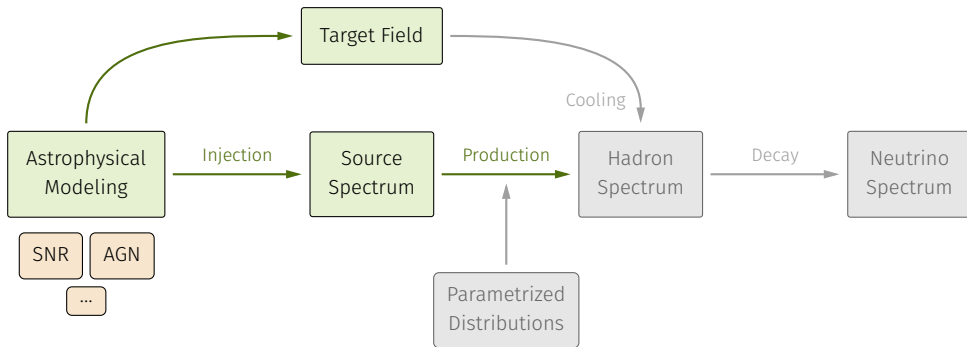


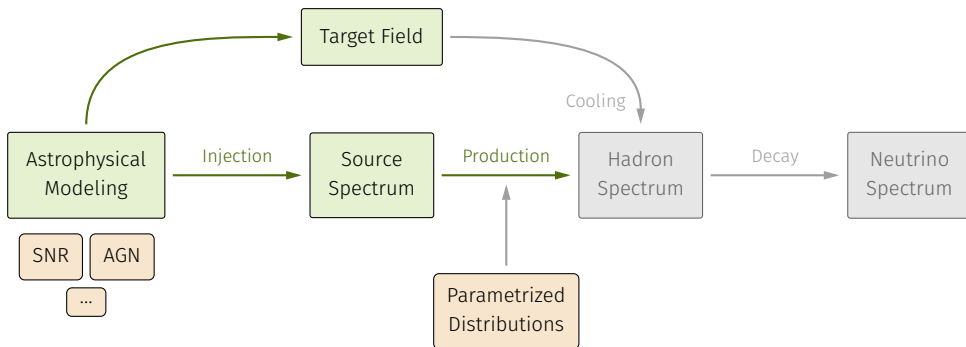


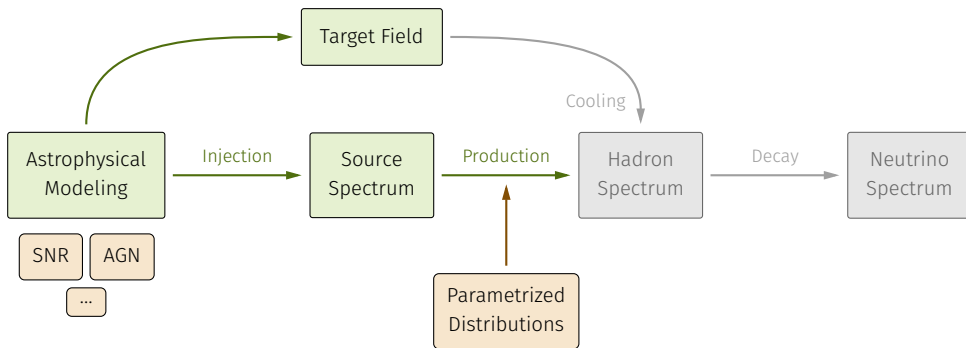




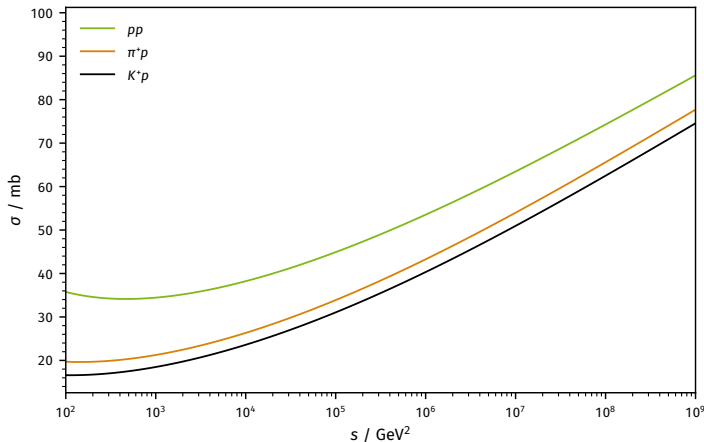








Hadron Inelastic Scattering Cross Section

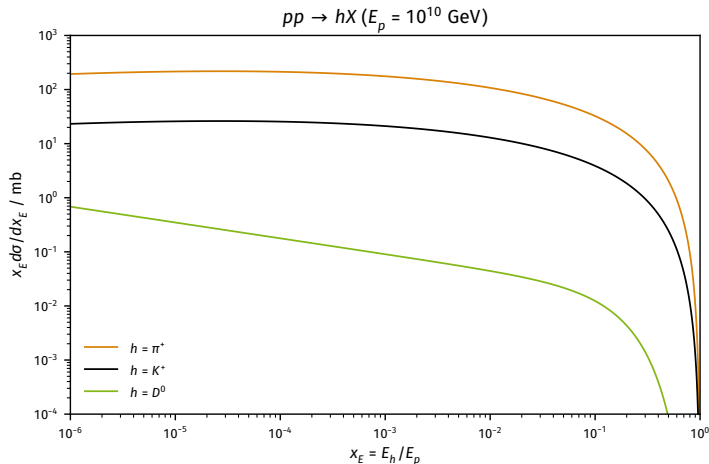


Combined Parametrizations

V. I. Belousov et al.
Phys. Atom. Nuclei (2016)

D. A. Fagundes, M. J. Menon
Nucl. Phys. A (2012)

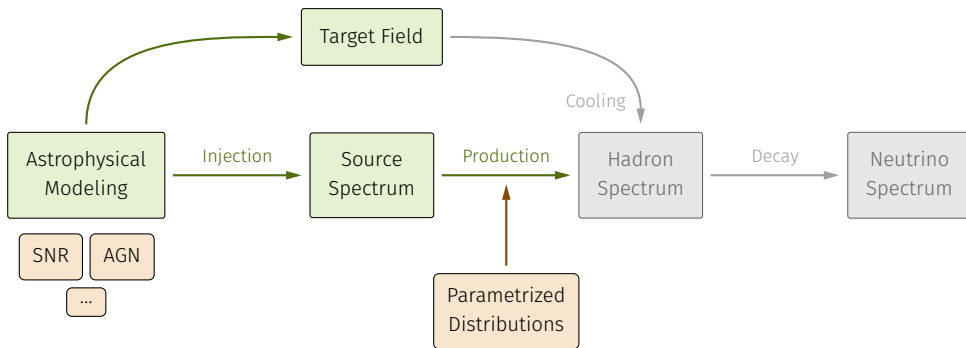
Weighted Energy Differential Cross Section

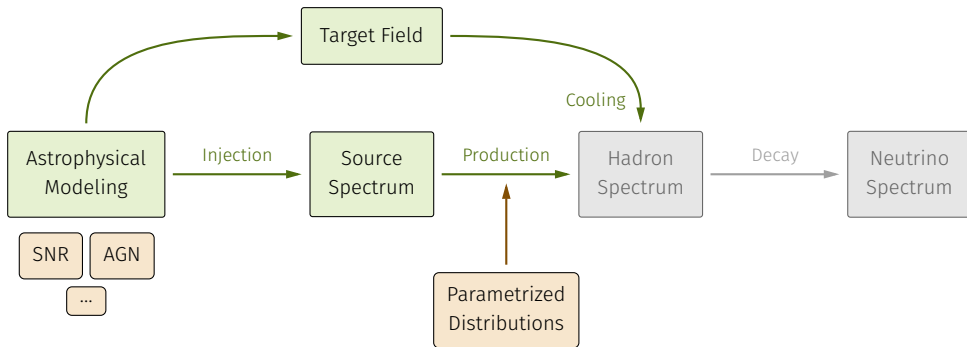


Derived Parametrizations

S. R. Kelner et al.
Phys. Rev. D (2006)

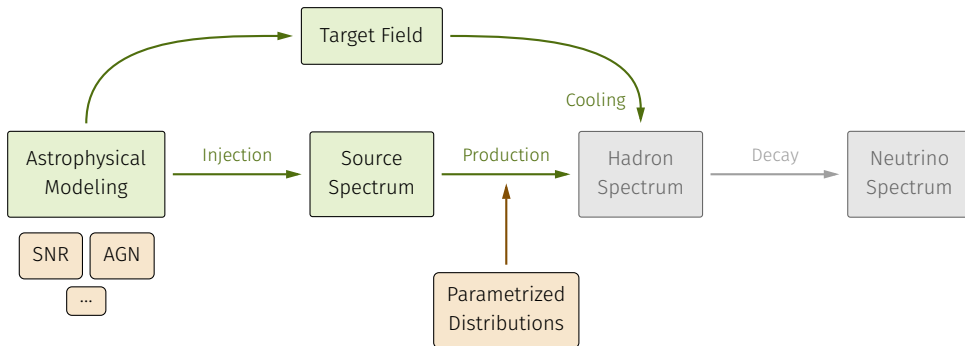
V. P. Gonçalves et al.
JHEP (2007)

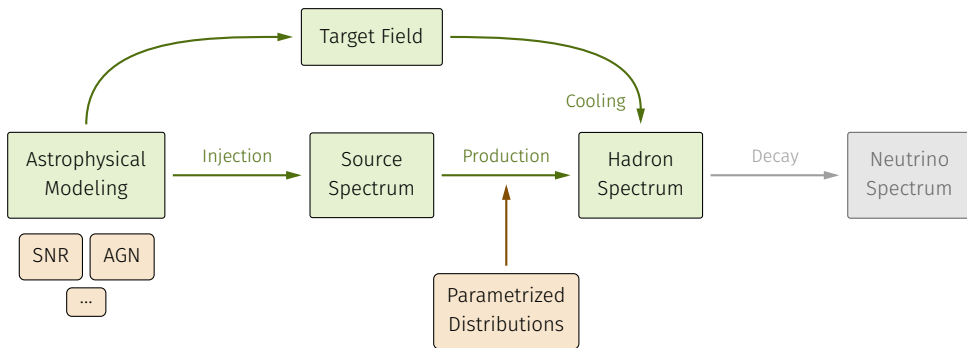




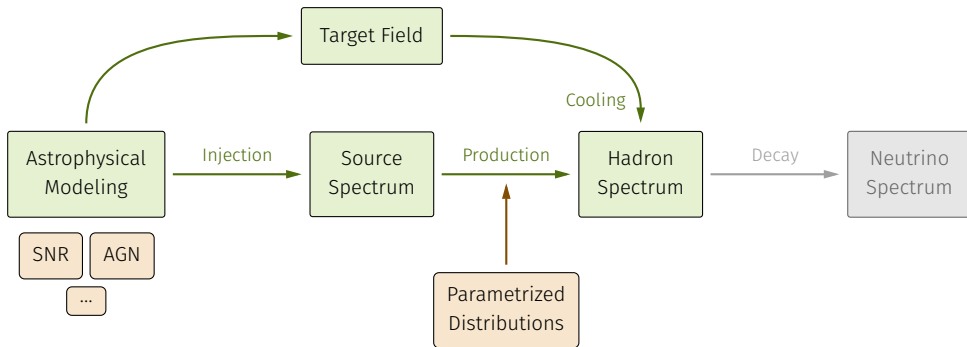


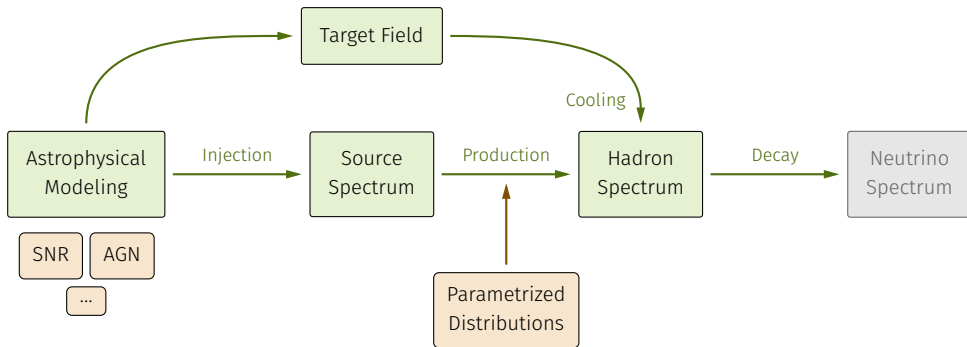
Cooling & Optical Depth



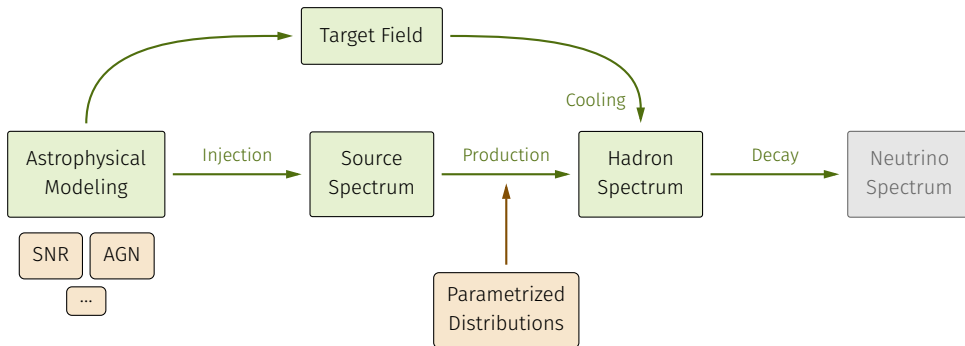


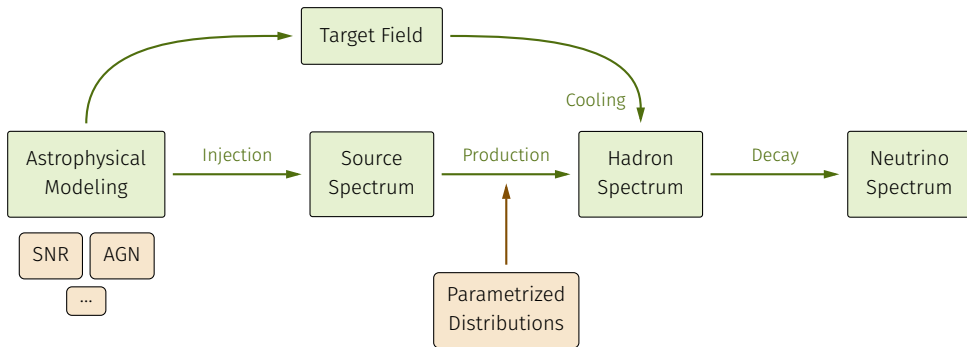




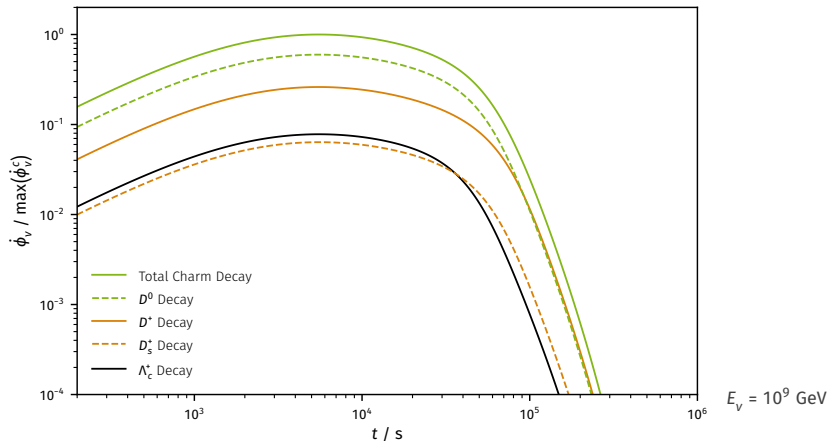




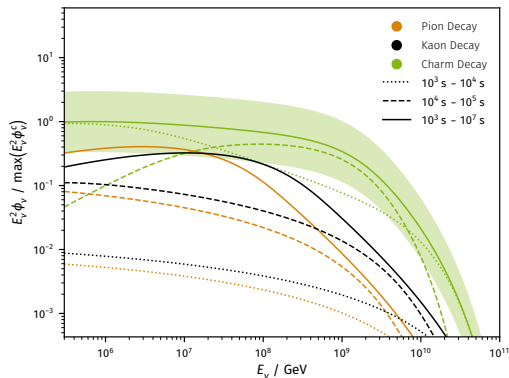
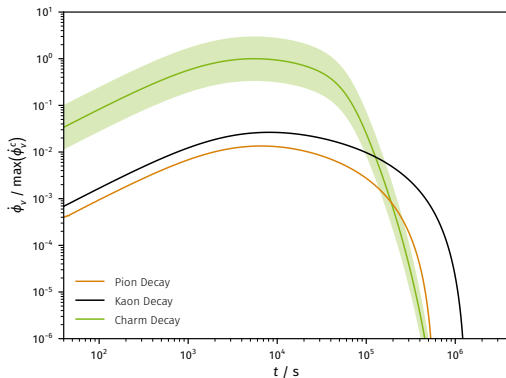




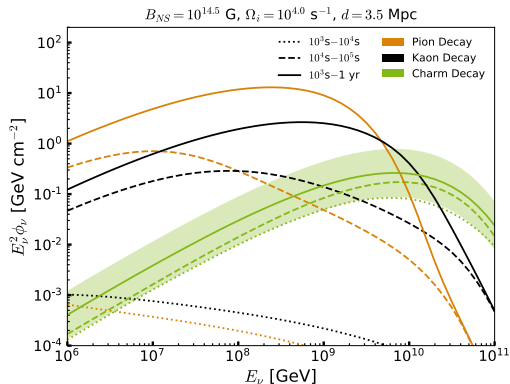
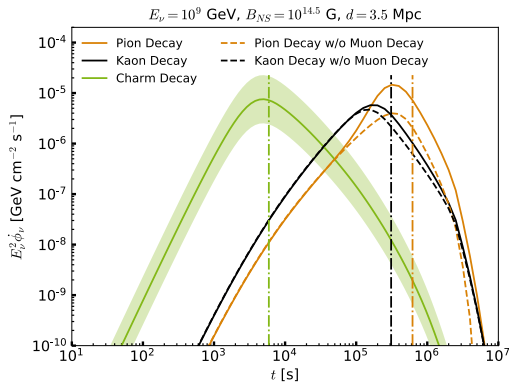
Normalized Magnetar Charm Contributions



Normalized Magnetar Neutrino Flux & Fluence

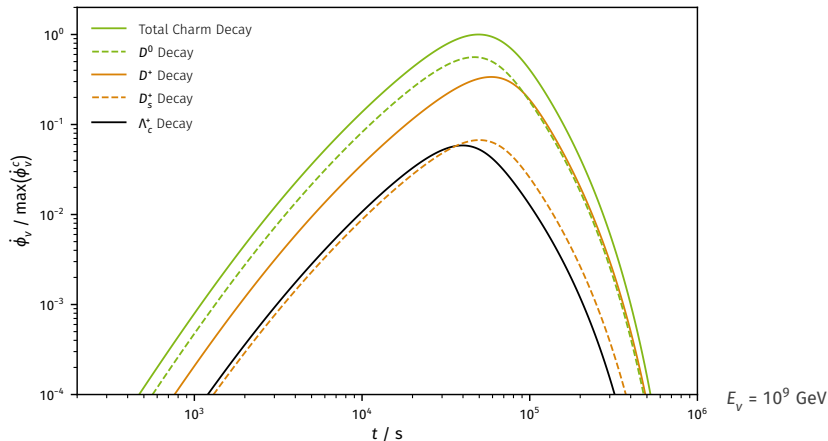


Comparison: Published Estimated Magnetar Neutrinos

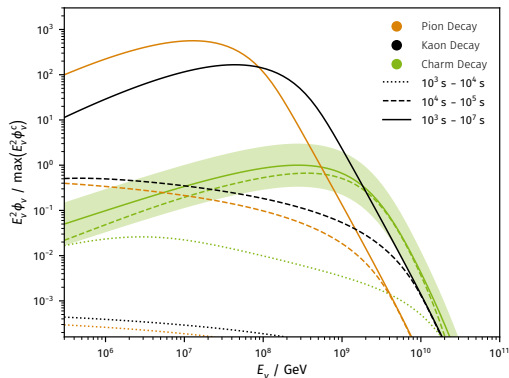
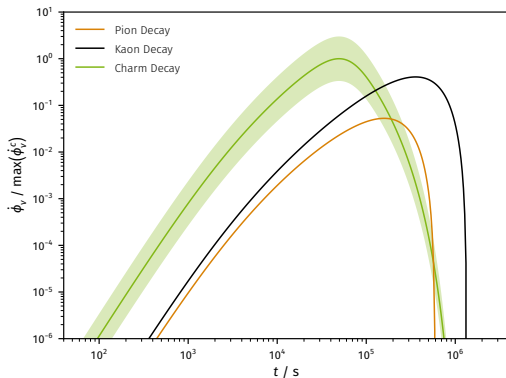


J. A. Carpio et al. *Phys. Rev. D* (2020)

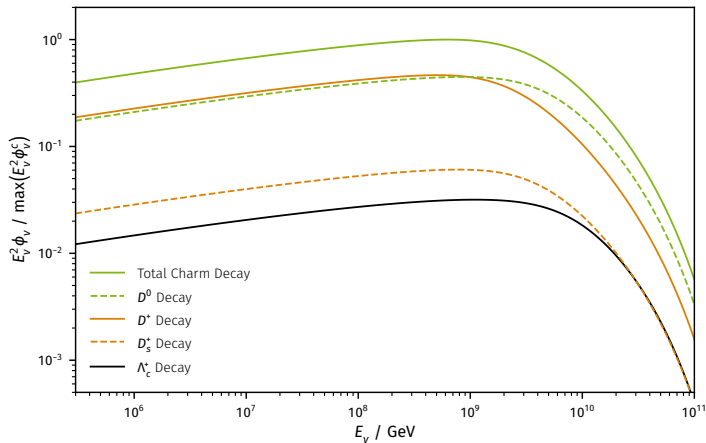
Modified Magnetar Charm Contributions



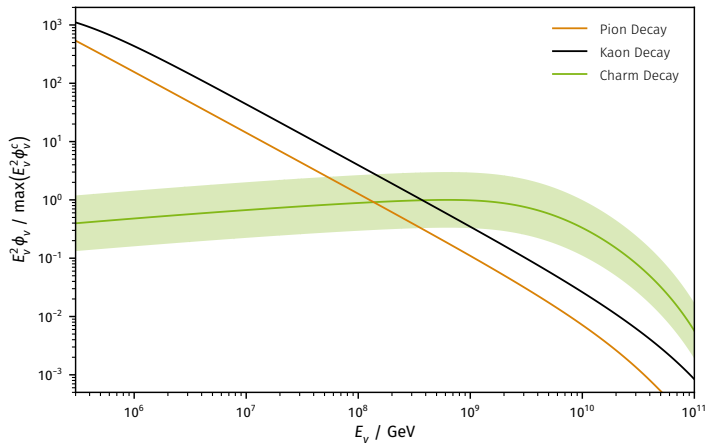
Modified Magnetar Neutrino Flux & Fluence



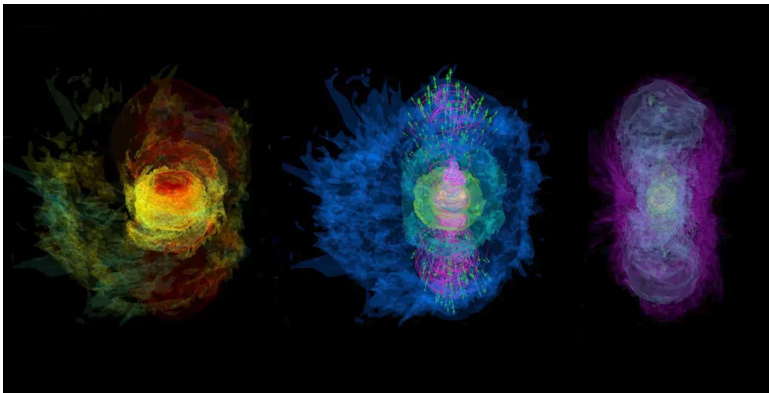
Normalized AGN Charm Contributions



Normalized AGN Neutrino Fluence

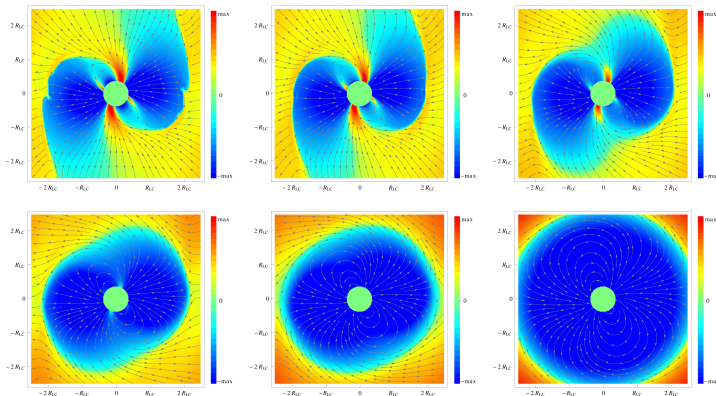


Simulated Merger Magnetar Formation



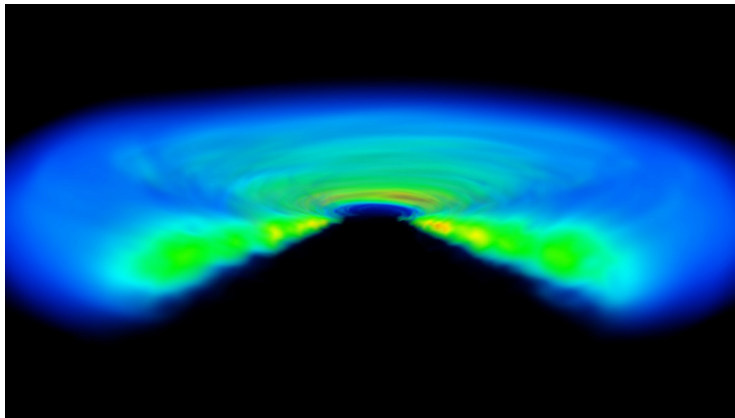
K. Hayashi *MPI Grav. Phys.* (2024)

Simulated Pulsar Magnetosphere



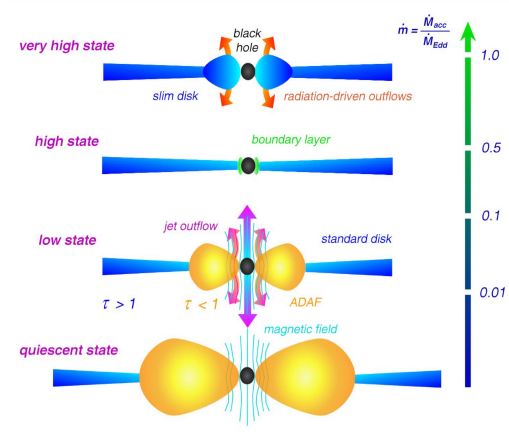
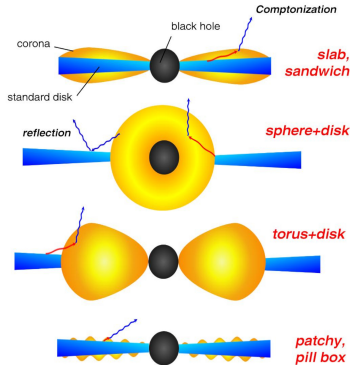
J. Li et al. *ApJ* (2012)

Simulated Accretion Disk Logarithmic Density

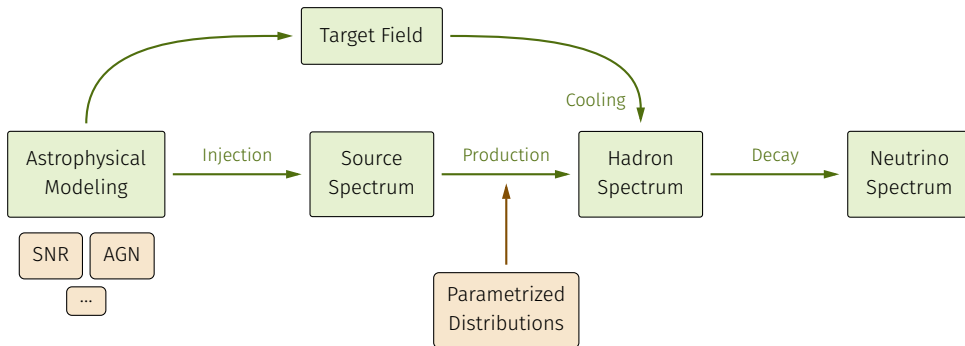


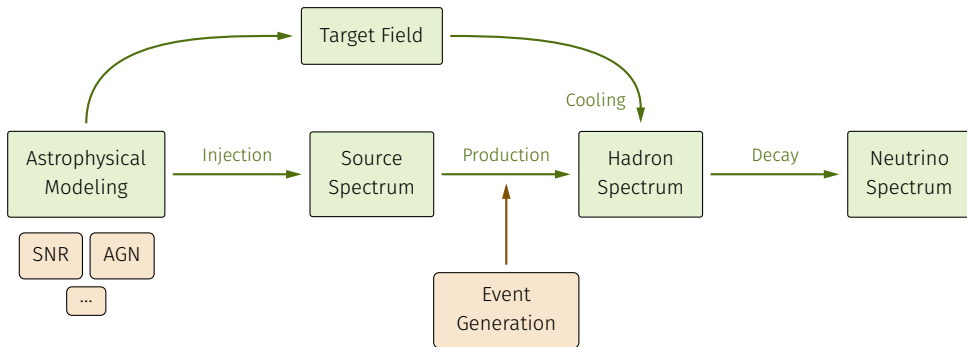
V. V. Zhuravlev et al. *ApJ* (2014)

AGN Accretion Disk & Corona

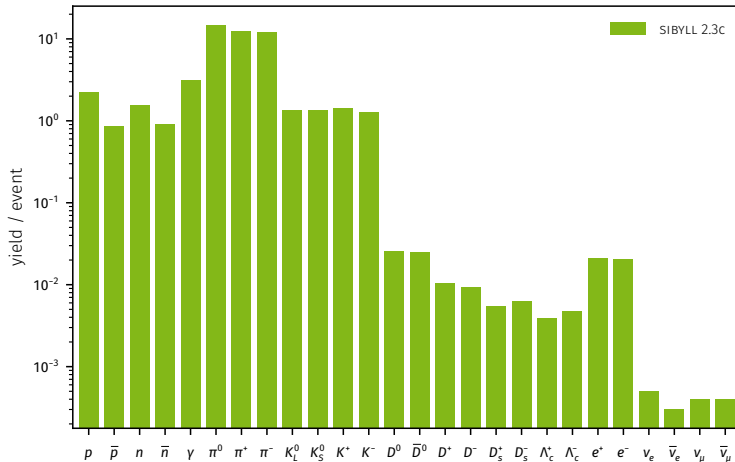


A. Müller University Heidelberg (2004)



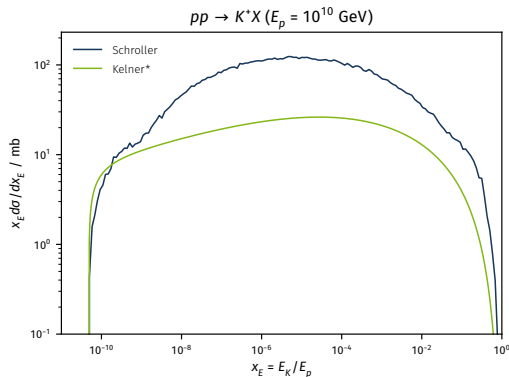
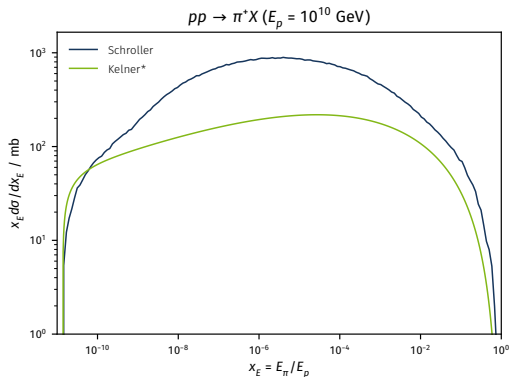


Monte Carlo Event Generation



Data Courtesy
M. Schroller

Comparison: Differential Cross Section



Derived Parametrization S. R. Kelner et al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schroll